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Assignment - 1

Machine learning

1) Machine learning to identify poverty - prove regions.

Problem :- Identify regions where people are at high risk of poverty.

Input data :- Income, employment, education, housing, population, healthcare access and living conditions.

Algorithm :- Decision Tree

Working :-

Socio - economic data
↓
Data Preprocessing
↓
Feature selection
↓
Decision tree
↓
Poverty risk classification
↓
High / Medium / low risk

The Decision tree divides regions using important socio economic factors. Regions with low income, unemployment & poor living conditions can be classified as poverty-prone.

Outcome :- Goverment can identify high risk regions and prioritize welfare programs.

... the efficient distribution

Advantages:- Easy to understand interpretable and useful for policy planning.

Limitations:- Poor-quality or incomplete socio-economic data can reduce accuracy.

Conclusion:- ML can support targeted poverty-reduction programs.

2) Identify eligible welfare-scheme beneficiaries:-

Problem:- Automatically identify people eligible for government welfare scheme.

Features:- Income, family size, employment rage, disability status, housing and existing benefits.

The model learns from previously verified applications and predicts eligibility for new applications.

Algorithm:-

Decision Tree
↓
Applicant Data
↓
Data cleaning
↓
Feature selection
↓
Decision tree model
↓
Eligibility prediction
↓
Eligible / Not eligible.

Outcomes:- Faster and more consistent identification of beneficiaries.

Advantages:- Reduces manual work and processing time.

Limitations:- Bias in historial applications data may produce unfair predictions

Conclusion:- ML can improve the efficient distribution of welfare benefits

3) Crop Yield prediction:-

Problem:- Predict crop yield using historical agricultural data

Features:- Rainfall, temperature, soil type, fertilizer, irrigation crop type and previous yield.

Algorithm:- Decision Tree Regression

Historical Agricultural Data

↓

Data preprocessing

↓

Feature selection

↓

Decision tree Regression

↓

Yield prediction

↓

Estimated crop yield

Outcomes:- Farmers can plan irrigation, fertilizers and harvesting

Advantages:- Supports agricultural planning and food-security decisions.

Limitations:- Unexpected weather conditions can affect predictions.

Conclusion:- ML-based yield prediction can contribute to food security.

4) Crop disease Detection using leaf Images

Problem:- Detect diseases from crop leaf images

Input:- Images of healthy and diseased leaves.

Algorithm:- Decision tree can be used with extracted image features, image-based system can also use classification techniques.

Leaf image

↓

Image preprocessing

↓

Feature extraction

↓

ML classifier

↓

Disease classification

↓

Health / Diseased

Outcomes:- Early disease identification

Advantages:- Helps farmers taste timely action

Limitations:- Poor lighting and image quality may affect prediction.

Conclusion:- ML can help reduce crop losses through early disease detection.

5) Reducing Food wastage in Supermarket.

Problem :- Predict product demand to prevent overstocking and wastage.

Features :- Sales history, season, price, holidays, product type and inventory

Algorithm :- Decision tree Regression

sales + Inventory Data
↓
preprocessing
↓
Feature selection
↓
ML model
↓
Demand prediction
↓
stock planning

Outcomes :- stores maintain appropriate inventory levels

Advantages :- Reduces food waste and improves inventory management.

Limitation :- Sudden changes in customer demand may reduce accuracy.

Conclusion :- ML can help supermarkets reduce unnecessary food wastage.